Wilson trialities: current corrected source contract

Human Compression project: L023 complete corrected contract

The fixed original is Dimitri Leemans and Klara Stokes, *Incidence geometries with trialities coming from maps with Wilson trialities*, arXiv:2208.08215v2, 9 September 2022, thirteen physical pages. This is a complete corrected principal submission for distinct assembly review, not a claim that all literal original assertions are true. All five complete written components are acquired, not treated as free citations:

4 pages	*Cubic field automorphisms and Wilson trialities*: root's exact native edition preserved in the distinct cubic-generation review.
3 pages	*Exhausting the cubic matrix equations*: root's frozen polynomial proof and complete required finite table program.
1 page	*A boundary case for the rank-four map construction*: root's digonal counterexample and its distinct review.
5 pages	*Flag completion and Wilson symmetry in the four-type construction*: new uniform proofs and exact small required certificate, with root's distinct bounded review.
3 pages	*Completing the odd projective-linear triality geometries*: the uniform all-flags proof, including its precise signed-vector scope.

The preliminary status sentences in these immutable native documents are historical. Current acceptance, authorship and limits are supplied by the exact review bindings, not by silent edits to those documents.

Principal results and limits

Source Theorem 3's odd-characteristic existence is proved by the complete subfield-counting argument: every odd prime power q admits a generating marked triple for $\mathrm{PSL}_2(q^3)$ with Wilson class III. The even-characteristic part is explicitly retained from Jones–Poulton [source reference 18, Theorem 2.3], as in the original; we do not claim a new proof of that classification/construction. The usual marked-map correspondence is retained. It is not an assertion that every abstract marked quotient is a faithful vertex–edge–face incidence map.

The polynomial component exhausts source equations (1)–(3), proves the primitive-eigenvalue version of Lemma 5 and the characteristic-dependent periods, and computes every one of the twelve rows of Table 1. All nine odd rows agree. The three even rows are corrected to $(24, 24)$, $(216, 144)$ and $(960, 1920)$ for $q = 4, 8, 16$ respectively. The additional solutions have $bc = 0$; this identifies the exact discrepancy, not the source authors' unpublished computational procedure. A primitive root in equation (4) is necessary for the stated large-order proof. The source's experimental remark concerning other orders is not an exhaustive new theorem and is not promoted to one here. If "root of unity" in the fourth equation is read without primitivity, there is an immediate literal counterexample to Lemma 5: in characteristic two take $R_0 = R_2 = I$ and $\xi = 1$. All four displayed equations hold, but the projective group generated

with $R_1 = J$ has order two. This does not dispute the primitive-root interpretation used for the corrected finite columns.

Construction 6 always defines its literal four-type coset pregeometry. The unconditional geometry claim, Lemma 7, is false even for a faithful Class III map: the complete product example has group order 16128 and a pairwise incident face–vertex–Petrie triple with no completing edge. The corrected local theorem proves two free G-orbits of chambers when the three nonedge parabolics are proper; an odd dihedron refutes unrestricted Lemma 8. Digonal and dihedral examples refute unrestricted Remark 9 and the Class I, II and IV correlation-quotient assertions.

The Class III symmetry assertion has a uniform faithful-map replacement. Local incidence valencies intrinsically fix the edge type, and restriction to the faithful original truncation excludes every type transposition. Thus the exact type image is C_3, even without assuming the marked group finite. The conclusion is about a pregeometry. For it to be an incidence geometry, the separate all-flags completion condition must hold. The explicit Class III counterexample proves that this qualification cannot be removed.

The principal infinite-family conclusion. The fifth component supplies the missing existential conclusion by a uniform written argument. For every odd prime power q, the accepted marking in $G = \mathrm{PSL}_2(q^3)$ has a three-dimensional adjoint model. A signed-vector stabilizer and norm argument bounds every face–Petrie residue by three vertices; three forced vertices and exactly two edges attain this bound. Thus every flag completes. Individual isotropic vectors are allowed, and injectivity is asserted only on a fixed face-star, not on the entire vertex set. The full automorphism group is G, the full correlation group is $G \rtimes C_3$, and there are two chamber orbits. Increasing q gives nonisomorphic geometries. This proves the advertised infinite existence while leaving the false unrestricted construction explicitly refuted.

The source's final broad question asks for infinitely many *flag-transitive* geometries with trialities and no dualities. It is answered by the separately accepted later Leemans–Stokes–Tranchida result (L015): for every odd $n > 1$ divisible by three, $\mathrm{Sz}(2^n)$ is the full group of such a thin, residually connected rank-three geometry, with correlation group $G \rtimes C_3$. The accepted generation repair is part of that attributed later result. This is not a claim of rank-four flag transitivity, not a premise of our odd-PSL completion, and not new question credit. Its exact theorem and acceptance are bound separately in the historical disposition record.

Borrowed structure and one printed parity correction

The entire subgroup classification in source Theorem 1 is borrowed Dickson–Vanden Cruyce material, not newly proved by this assembly. The actual structural inputs used by the new proofs are the subgroup types, canonical subfield normalizers, Klein normalizers and involution centralizers, the standard projective semilinear automorphism group, and finite-field matrix Hilbert 90. Their complete external proof and software costs are unknown. The cubic proof supplies the new uniform bound on obstructing subfields rather than relying on the source's claim that all proper stable generated groups lie in the cubic subfield.

Source Lemma 2 remains valid in its surrounding odd-q context. The cube of an odd residue is the same residue modulo eight. Thus the retained Klein-normalizer classification gives the same order (12 or 24) in $\mathrm{PSL}_2(q)$ and $\mathrm{PSL}_2(q^3)$. The former normalizer is contained in the latter; their equal orders prove equality. We do not extend this assertion to even characteristic.

For precision, one literal parameter in Theorem 1(6), physical page 6, is reversed. Let p be odd, $k \mid r$. An element of $\mathbb{F}_{p^k}^*$ is a square in $\mathbb{F}_{p^r}^*$ according to its square character raised to

$$1 + p^k + \cdots + p^{r-k} \equiv r/k \pmod 2.$$

Consequently

$$\left|\mathbb{F}_{p^k}^* \cap (\mathbb{F}_{p^r}^*)^2\right| = \begin{cases} p^k - 1, & r/k \text{ even}, \\ (p^k - 1)/2, & r/k \text{ odd}. \end{cases}$$

These are the possible Borel multipliers in the ambient projective special linear group. The source assigns the denominator 2 to the even extension and 1 to the odd extension, in the opposite order. For example its $q = 3$, $s = k = r = 1$ entry permits $E_3 : C_2$, which would be a subgroup of order six in $\mathrm{PSL}_2(3) = A_4$. There is no such subgroup: an index-two subgroup would yield a nontrivial map $A_4 \to C_2$, whereas A_4 is generated by its elements of order three. This is an explicit ancillary correction, not a change to any accepted component's hypotheses.

Complete acquisition and attribution

The source existence result and matrix equations are credited to Leemans and Stokes, the even family to Jones and Poulton, and the retained classification and map correspondence to the named external inputs. Root authored the four-page cubic, three-page polynomial and one-page digonal components. The present author distinctly reviewed the cubic and digonal proofs and authored the five-page symmetry/counterexample component and the new three-page completion proof. Root supplied the symmetry component's separate distinct review and scrutinized the completion proof. The polynomial component has its own distinct review. No author self-accepts a new proof.

The complete `polynomial_table.g` is required for the twelve finite rows; the complete `certificate.py` is required for the displayed fixed $\mathrm{SL}_2(8)$ instance. Their recorded output is regenerated evidence, not an additional mathematical database. All operational drivers and independent optional diagnostics are separately inventoried in full. No new Lean declaration, source-count admission, new question credit or numerical factor is supplied. The source mentions Magma but supplies no complete executable listing; its software cost is unknown, not zero. Any later measurement must preserve the user's symmetric treatment of executable verification.